

V Muhammed Saad Sabeel

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EDUCATION

BMS Institute of Technology & Management

Bachelor of Engineering in Electronics & Telecommunication Engineering, CGPA: 9.24

Bengaluru, India

2023 – 2027

Aragami PU College

Class 12 Science, Percentage: 89%

Bengaluru, India

2022 – 2023

PROJECTS

Automated Offline-to-Digital Operations Pipeline | *Python, OCR, Google Sheets API*

- Engineered a computer vision automation pipeline using Python and OCR (OpenCV) to extract unstructured handwritten text from physical forms, eliminating manual data entry.
- Designed a data validation script to clean and standardize extracted text before automatically injecting the data into a Google Sheets database via API.
- Optimized operational workflows, reducing the manual turnaround time for vendor onboarding document processing to under few seconds per form.

Automated Manufacturing KPI Scorecard | *Python, Pandas, Power BI, DAX*

- Engineered an automated ETL pipeline using Python to extract, clean, and transform unstructured daily production logs into an analysis-ready format.
- Built an interactive Power BI dashboard with dynamic DAX measures to track real-time machine metrics, isolate anomalies, and eliminate manual reporting.

Diabreeze: Non-Invasive Biomarker Detection System | *ESP32, Data Processing*

- Developed a portable diagnostic system that extracts and processes real-time breath acetone telemetry as a biomarker to estimate blood glucose levels.
- Engineered an ESP32 microcontroller data pipeline to compute and visualize physiological metrics instantly.

QNX Deterministic Security Monitor | *C, QNX RTOS, Microkernel Architecture, IPC*

- Designed a split-architecture security solution on a QNX Microkernel using strictly isolated C programs to mitigate hardware side-channel attacks.
- Developed a zero-overhead Mitigation Engine (Priority 63) triggered by native IPCs to instantly preempt and freeze compromised processes mid-instruction, preventing data exfiltration.

IoT Intrusion Detection via Side-Channel Analysis | *Raspberry Pi, I2C, Python, Edge AI*

- Building a non-intrusive edge monitoring system utilizing hardware power side-channel analysis to detect malware, bypassing the need for software logs.
- Engineered an I2C data acquisition pipeline to sample current consumption every 0.2 seconds, generating a custom dataset to train real-time Edge AI classification models.

EXTRACURRICULAR

Web Development Intern | *Bhisma's Aero Pvt. Ltd., Bengaluru*

- Developed and maintained web applications for aerospace industry client requirements.

AI Model Evaluator | *Outlier*

- Contributed 330+ hours of RLHF and SFT tasks to train enterprise LLMs, evaluating model outputs for factual accuracy and logical consistency.
- Acted as an Oracle-tier grader across Mathematics and multilingual (English/Hindi) domains, applying strict quality assurance protocols to AI-generated data.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Basics of C, MATLAB

Databases & Analytics: PostgreSQL, Fundamentals of Power BI

Tools & Software: Git, VS Code, Cisco Packet Tracer, OpenCV